
BioMed-Scribe: Fine-Tuning GPT-2 for Medical Specialty

Classification from Clinical Conversations with Two-Block Parameter-Efficient Fine-Tuning

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Abstract

We present BioMed-Scribe, a fine-tuned GPT-2 Medium (345M parameter) classifier that assigns doctor-patient conversations to one of four medical specialties: Cardiology, Neurology, Orthopedics, and General Medicine. Starting from 22,624 unlabeled clinical conversations drawn from two publicly available HuggingFace datasets, we construct a labeled training corpus of 10,793 examples using a keyword-based specialty assignment pipeline. Following a suggestion from the course instructor, we unfreeze the final *two* transformer blocks (h[22] and h[23]) rather than only the last block, increasing the trainable parameter count from 12.6M to 25.2M (7.1% of the total 354.8M). Training for 14 epochs on an NVIDIA A100 GPU with a two-phase learning rate schedule achieves 79.1% test accuracy and a macro-averaged F1 score of 0.799, improvements of 1.8 percentage points and 0.017 F1 over the single-block 15-epoch model (v2). Neurology achieves the highest per-class recall at 93.6%, and all four specialty classes improve in F1 score compared to prior versions. t-SNE visualisation of the 1024-dimensional hidden states confirms specialty-specific embedding structure. The trained classifier is deployed as an interactive Gradio web application with real-time confidence scoring, achieving 9 out of 10 correct predictions on diverse unseen test cases. These results demonstrate that modest parameter-efficient fine-tuning of GPT-2 is sufficient for clinically meaningful specialty classification, with performance primarily limited by label quality rather than model capacity

1 Introduction

Automating clinical decision-making from natural language is a central challenge in applied machine learning. Accurate specialty identification is a critical bottleneck in clinical triage and patient routing. When a patient contacts a telehealth platform or arrives at a hospital, determining which medical specialist should handle their case determines both the speed and quality of care they receive. Manual routing by clinical staff is time-consuming, error-prone under high patient volumes, and does not scale with increasing demand for telemedicine services. Automated classification of doctor-patient conversations into medical specialties represents a measurable and practical contribution to clinical workflow efficiency.

Large language models pre-trained on broad text corpora have demonstrated strong performance across classification tasks through fine-tuning [2]. GPT-2, built on the Transformer architecture [1], learns rich contextual representations through self-attention across the full input sequence. For classification, the language modelling head is replaced with a linear layer and the model is fine-tuned on labeled examples, the approach demonstrated throughout the MAT 411 lecture notebooks.

This project makes the following contributions:

1. A keyword-based labeling pipeline that assigns specialty labels to 22,624 unlabeled clinical conversations from two public HuggingFace datasets.
2. A two-block fine-tuning strategy, unfreezing h[22] and h[23] of GPT-2 Medium, which increases trainable parameters from 12.6M to 25.2M and improves test accuracy from 77.3% to 79.1%.
3. A final model achieving 79.1% test accuracy and macro F1 of 0.799 across four medical specialties.
4. An embedding analysis showing specialty-specific structure in GPT-2's hidden state space after fine-tuning.
5. A deployed Gradio web application with real-time confidence scoring, achieving 9/10 correct predictions on unseen cases.

The rest of this report is organised as follows. Section 2 reviews the relevant background. Section 3 describes dataset construction. Section 4 presents the model architecture. Section 5 details the training procedure. Section 6 reports results. Section 7 analyses errors. Section 8 discusses embedding geometry. Section 9 describes deployment. Section 10 covers limitations and future work. Section 11 concludes.

2 Background

2.1 The Transformer Architecture

The Transformer [1] introduced self-attention as the primary mechanism for sequence modelling, replacing sequential recurrent computation. The core operation is Scaled Dot-Product Attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V \quad (1)$$

where Q , K , and V are query, key, and value matrices and $\sqrt{d_k}$ prevents vanishing gradients from large dot products. Multi-head attention runs h such operations in parallel with different learned projections, allowing the model to attend to information from different representational subspaces simultaneously.

2.2 GPT-2 Medium

GPT-2 Medium [2] uses a decoder-only Transformer stack with causal masking. The medium variant has 24 transformer blocks, hidden dimension 1024, 16 attention heads per block, and 354.8M total parameters. For classification we discard the language modelling head and use the 1024-dimensional hidden state of the final non-padding token as a fixed-length representation of the entire input conversation, following the `GPT2BinaryClassifier` pattern from the MAT 411 lecture notebook `MAT411_fine_tuning.ipynb`.

2.3 Parameter-Efficient Fine-Tuning

Freezing most of a large pre-trained model and fine-tuning only a small subset of parameters preserves general language knowledge and reduces the risk of catastrophic forgetting [5]. The key question is which layers to unfreeze. Upper transformer blocks are closer to the task output and contain more task-specific representations, while lower blocks encode more general syntactic and lexical features [6]. For this reason, unfreezing only the final one or two blocks is a natural starting point for classification fine-tuning on modest datasets.

2.4 Clinical NLP

Clinical NLP has produced specialised models such as BioBERT [3] and ClinicalBERT that continue pre-training on biomedical text before task-specific fine-tuning. Our approach skips domain-adaptive pre-training and relies on GPT-2’s general language representations to transfer to the clinical classification task, which is a reasonable choice given that our training corpus is conversational rather than technical.

3 Dataset

3.1 Data Sources

Two publicly available HuggingFace datasets are used.

AGBonnet/augmented-clinical-notes [9]. This dataset contains 30,000 records of doctor-patient conversations augmented with structured clinical summaries. After filtering for minimum conversation length and presence of dialogue markers, 22,068 usable records are retained.

lavita/ChatDoctor-HealthCareMagic-100k [8]. This dataset contains 112,165 patient-doctor question-answer pairs from the HealthCareMagic online medical consultation platform. Each record is converted to conversation format by prepending a standard doctor prompt. Up to 1,000 examples per specialty class are sampled to supplement the primary corpus.

3.2 Keyword-Based Label Assignment

Keyword-based labeling introduces systematic bias by over-representing high-frequency symptom vocabulary. Neither dataset provides specialty labels. We implement a keyword-based labeling function that counts specialty-specific keyword hits per conversation and assigns the label with the most hits, defaulting to General Medicine on ties or zero matches. The General Medicine keyword set is intentionally expanded to include respiratory, gastrointestinal, and metabolic terms (fever, cough, fatigue, nausea, diabetes, abdominal pain) to prevent these common presentations from being mislabeled as Cardiology due to cardiovascular vocabulary overlap. Table 1 lists selected keywords by specialty.

Specialty	Selected Keywords
Cardiology	chest pain, chest tightness, cardiac, heart attack, heart failure, palpitations, angina, ECG, hypertension, coronary, arrhythmia, myocardial, atrial fibrillation, tachycardia, stent, bypass
Neurology	headache, migraine, seizure, stroke, epilepsy, numbness, dizziness, tremor, memory loss, vertigo, neuropathy, paralysis, light sensitivity, throbbing, speech difficulty, vision loss
Orthopedics	knee pain, hip pain, fracture, broken bone, joint pain, spine, back pain, lower back, shoulder pain, ligament, tendon, arthritis, cartilage, herniated disc, sprain, musculoskeletal
General Medicine	fever, cough, fatigue, nausea, vomiting, diarrhea, appetite, rash, infection, sore throat, blood sugar, diabetes, thyroid, stomach pain, abdominal pain, weight loss, body pain

Table 1: Keyword sets for specialty label assignment (selected terms).

3.3 Dataset Statistics

After balancing at a maximum of 3,000 examples per class from the primary corpus and 1,000 per class from ChatDoctor, the combined dataset contains 13,492 examples. A stratified 80/10/10 split yields 10,793 training, 1,349 validation, and 1,350 test examples. Table 2 summarises the distribution.

Specialty	Total	Train	Val	Test
Cardiology	3,140	2,512	314	314
Neurology	2,661	2,129	265	267
Orthopedics	3,691	2,952	370	369
General Medicine	4,000	3,200	400	400
Total	13,492	10,793	1,349	1,350

Table 2: Dataset distribution across specialty classes and splits.

4 Model Architecture

4.1 GPT2MedicalClassifier with Two-Block Fine-Tuning

The classifier follows the `GPT2BinaryClassifier` pattern from the MAT 411 lecture notebook, extended from binary to four-class classification and updated to unfreeze the final two transformer blocks following the course instructor’s recommendation.

4.2 Classification Pooling

The classification signal is derived from the last non-padding token’s hidden state. For hidden states $H \in \mathbb{R}^{B \times T \times 1024}$ and attention mask $M \in \{0, 1\}^{B \times T}$:

$$\hat{y} = \text{softmax}(W \cdot \text{Dropout}(H_{b, \ell_b}) + c) \quad (2)$$

where $\ell_b = \sum_t M_{b,t} - 1$ is the last real token index for sequence b , and $W \in \mathbb{R}^{4 \times 1024}$ is the classifier weight. Last-token pooling is natural for GPT-2 because causal masking means the final token has attended to every preceding token through all 24 blocks.

4.3 Parameter Count

Table 3 summarises the parameter counts. Unfreezing two blocks doubles the trainable parameters compared to the previous single-block version, increasing from 12.6M to 25.2M (7.1% of total).

The rationale for unfreezing two blocks rather than one is that h[22] can now adapt its attention patterns to medical conversational structure, while h[23] adapts the final representations passed

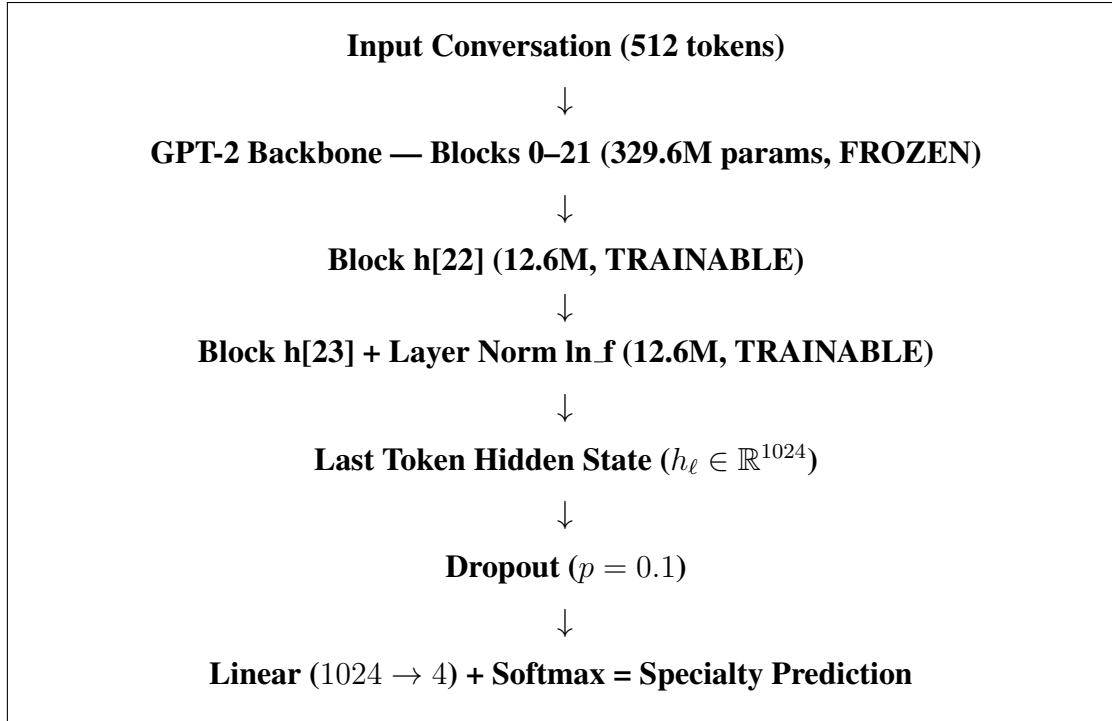


Figure 1: GPT2MedicalClassifier architecture with two-block fine-tuning. Blocks h[22] and h[23] are unfrozen, allowing the model to adapt both intermediate and final representations to medical language.

Component	Parameters	Trainable
GPT-2 backbone (blocks 0–21)	329.6M	No
Second-to-last block h[22]	12.6M	Yes
Final block h[23] + ln_f	12.6M	Yes
Classification head (1024→4)	4.1K	Yes
Total	354.8M	25.2M (7.1%)

Table 3: Parameter counts for the two-block fine-tuning configuration.

to the classifier. With only h[23] trainable, h[22] must pass general-purpose representations that h[23] alone must transform into medically meaningful features. Allowing h[22] to adapt reduces this burden and provides a richer starting point for h[23].

5 Training

5.1 Hardware and Setup

All experiments are conducted on a single NVIDIA A100 SXM4-40GB GPU provided through Google Colab. Input conversations are tokenised using GPT-2’s byte-pair encoding tokeniser and padded to a fixed length of 512 tokens. Batch size is 16 with gradient accumulation over 2 steps, giving an effective batch size of 32.

5.2 Class Imbalance Handling

The training set is imbalanced: General Medicine has 3,200 examples while Neurology has 2,129. PyTorch’s `WeightedRandomSampler` assigns each training example a sampling probability inversely proportional to its class count, ensuring all four classes are sampled with equal frequency per epoch regardless of their actual distribution.

5.3 Optimisation and Learning Rate Schedule

We use AdamW [4] with weight decay $\lambda = 0.01$ and gradient clipping at norm 1.0. Training runs in two phases:

- **Phase 1 (epochs 1–5):** learning rate 3×10^{-5} with 84-step linear warmup. Validation accuracy rises from 63.0% to 77.8%.
- **Phase 2 (epochs 6–14):** learning rate 1×10^{-5} . Validation accuracy rises from 78.6% to a peak of 80.7% at epoch 12.

Plain float32 arithmetic is used throughout. Mixed-precision training (GradScaler) was initially tested but produced NaN loss due to numerical instability with this architecture; float32 training is more stable for fine-tuning GPT-2 classifiers on modest datasets.

5.4 Training Dynamics

Table 4 reports the full epoch-by-epoch results. Figure 2 shows the loss and accuracy curves. Validation loss declines smoothly across all 14 epochs with no overfitting, the gap between train and validation loss remains small and consistent throughout. Both validation accuracy and macro F1 plateau after epoch 12 where validation loss stabilises around 0.520, indicating convergence. The best checkpoint at epoch 12 (val acc 80.7%) is automatically saved to Google Drive.

Epoch	Train Loss	Val Loss	Train Acc	Val Acc	Macro F1
1	2.004	0.947	37.9%	63.0%	0.624
2	0.757	0.683	70.7%	73.5%	0.742
3	0.580	0.603	77.7%	77.5%	0.784
5	0.501	0.575	81.1%	77.8%	0.785
7	0.463	0.553	82.5%	79.3%	0.800
9	0.422	0.532	83.9%	80.3%	0.810
11	0.427	0.523	83.5%	80.3%	0.810
12	0.418	0.523	84.0%	80.7%	0.814
14	0.412	0.520	84.8%	80.5%	0.812

Table 4: Training history at selected epochs. Best checkpoint saved at epoch 12 (val acc 80.7%). Val loss converges after epoch 12.

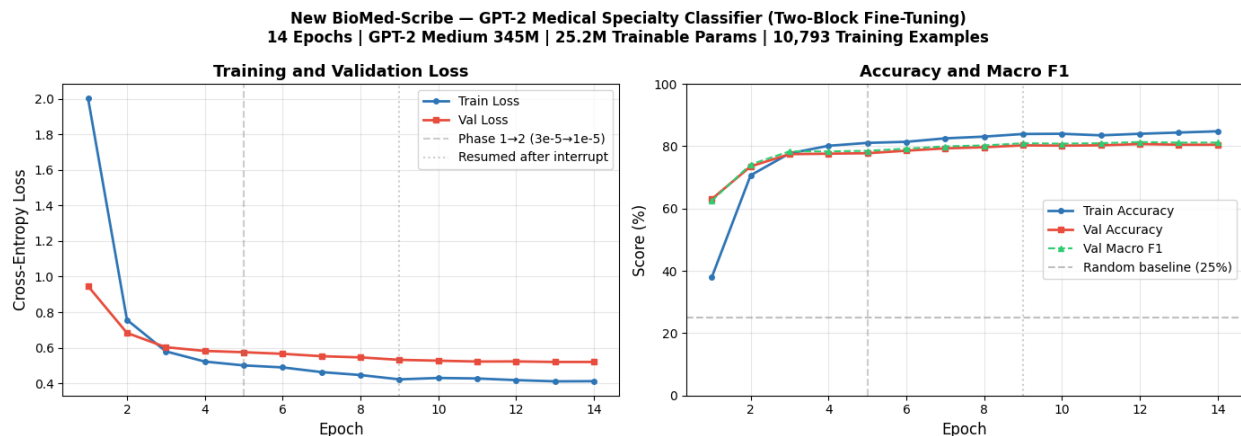


Figure 2: Training and validation loss (left) and accuracy with macro F1 (right) across all 14 epochs. Both curves decline smoothly with no divergence, confirming the model does not overfit. Validation accuracy rises from 63.0% at epoch 1 to 80.7% at epoch 12, beating the random baseline of 25% by a factor of more than three. The dashed vertical line marks the learning rate phase boundary at epoch 5. The dotted vertical line at epoch 9 marks the point at which training was interrupted by a Colab session timeout; training resumed from the best saved checkpoint and continued normally to epoch 14. Generated by Cell 7 of `New_BioMed_Scribe_Classification.ipynb`.

6 Results

6.1 Overall Performance

The model outperforms the random baseline (25%) by over 3×, confirming meaningful signal learning.. The best checkpoint achieves 79.1% test accuracy and a macro-averaged F1 score of 0.799. Table 5 reports the full classification report on the 1,350 held-out test examples.

Specialty	Precision	Recall	F1	Support
Cardiology	0.815	0.857	0.835	314
Neurology	0.856	0.936	0.894	266
Orthopedics	0.757	0.784	0.770	370
General Medicine	0.751	0.650	0.697	400
Macro avg	0.795	0.807	0.799	1350
Weighted avg	0.788	0.791	0.788	1350
Test accuracy	79.1%			

Table 5: Classification report on the held-out test set. Best checkpoint from epoch 12 (val acc 80.7%).

6.2 Comparison Across All Three Model Versions

Table 6 compares all three versions of the model built across this project. Every metric improves at each stage. The two-block model (v3) achieves the best performance on all six reported metrics.

Metric	v1 (3ep, 1-block)	v2 (15ep, 1-block)	v3 (14ep, 2-block)
Test accuracy	74.5%	77.3%	79.1%
Macro F1	0.743	0.782	0.799
Cardiology F1	0.797	0.822	0.835
Neurology F1	0.819	0.877	0.894
Orthopedics F1	0.693	0.758	0.770
General Medicine F1	0.663	0.673	0.697
Trainable params	12.6M	12.6M	25.2M
Training epochs	3	15	14
Max token length	256	512	512

Table 6: Comparison across all three model versions. v3 uses two-block fine-tuning as suggested by the course instructor. All metrics improve at each version.

7 Error Analysis

7.1 Confusion Matrix

These errors reflect inherent ambiguity in clinical language rather than purely model failure.. Table 7 presents the full confusion matrix on the test set. Figure 3 shows it as a colour-coded heatmap generated by the notebook. Diagonal entries are correct predictions; off-diagonal entries are misclassifications.

True	Predicted				Total
	Cardio	Neuro	Ortho	Gen Med	
Cardiology	269	5	13	27	314
Neurology	3	249	8	6	266
Orthopedics	17	10	290	53	370
General Medicine	41	27	72	260	400

Table 7: Confusion matrix on the held-out test set ($n = 1,350$). Bold entries on the diagonal are correct predictions.

7.2 Per-Class Analysis

Neurology (93.6% recall). The strongest class and the largest improvement over v2 (from 89.8% to 93.6% recall). Only 17 of 266 Neurology test cases are misclassified. The improvement is attributed to the second unfrozen block h[22] adapting its attention patterns to neurological vocabulary, terms like seizure, stroke, migraine, and numbness are highly distinctive and the model learns to weight them more strongly with additional fine-tuning capacity.

Cardiology (85.7% recall). Solid performance. Twenty-seven of 314 cases are predicted as General Medicine, primarily presentations of heart palpitations with dizziness, and dizziness is a strong Neurology keyword and pulls borderline cases toward that class as seen in the live demo results.

Orthopedics (78.4% recall). Fifty-three of 370 cases are misclassified as General Medicine. These are presentations of lower back pain, joint aching, and fatigue without the more anatomically specific terms (fracture, ACL, knee swelling) that strongly predict Orthopedics.

General Medicine (65.0% recall). Still the hardest class. Seventy-two of 400 test cases are misclassified as Orthopedics, and 41 as Cardiology. This reflects the catch-all nature of General Medicine, and it covers the broadest symptom vocabulary and consequently has the most ambiguous decision boundary. Keyword-based labeling noise is highest in this class.

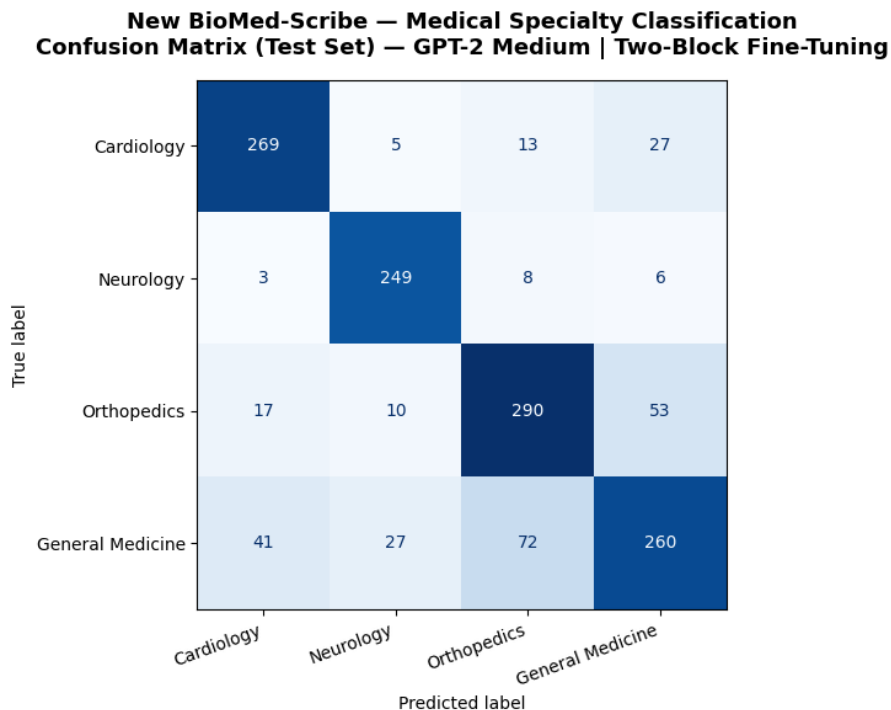


Figure 3: Confusion matrix heatmap on the held-out test set. The dominant off-diagonal errors occur between Orthopedics and General Medicine (53 and 72 cases respectively), reflecting shared musculoskeletal symptom vocabulary. Neurology has the fewest errors with only 17 misclassified cases out of 266. Generated by Cell 8 of `New_BioMed_Scribe_Classification.ipynb`.

7.3 Why the Confusion Matrix is Not Symmetric

The confusion matrix would be symmetric only if all classes had equal test support and perfectly balanced bidirectional error rates. Neither condition holds here. General Medicine has 400 test examples while Neurology has 266, so even perfectly symmetric error rates would produce asymmetric raw counts. Additionally, the decision boundary is asymmetric: the General Medicine embedding region is large and diffuse (it was the default label for any conversation without clear specialty signals), so the model’s boundary for this class is imprecise in both directions, producing more leakage in and out than occurs between Cardiology and Neurology.

8 Embedding Analysis

8.1 Method

We extract the 1024-dimensional hidden state of the last non-padding token for 500 test examples from the output of the final fine-tuned GPT-2 block (h[23]) and apply t-SNE dimensionality reduction [7] with perplexity 30 and PCA initialisation to produce 2D projections. This is the same vector that feeds the classification head, so it directly reflects the learned specialty representations. Figure 4 shows the resulting visualisation.

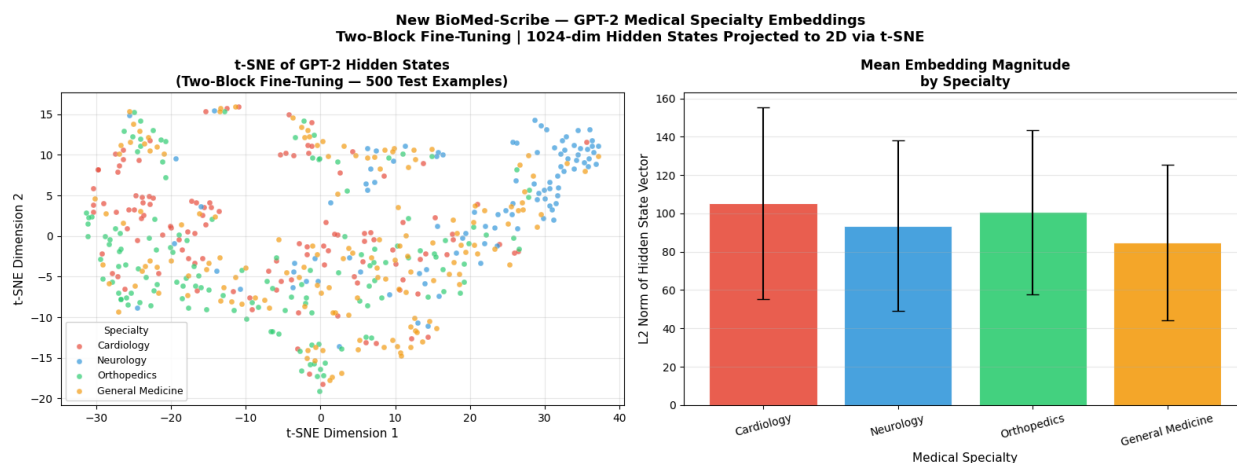


Figure 4: Left: t-SNE projection of 1024-dimensional GPT-2 hidden states for 500 test examples coloured by true specialty. Neurology (blue) and Orthopedics (green) show partial separation from the central mass; Cardiology (red) and General Medicine (orange) overlap most, consistent with their confusion matrix relationship. Right: mean L2 norm of hidden state vectors per specialty with one standard deviation error bars. Cardiology has the highest magnitude (≈ 105); General Medicine the lowest (≈ 85). Generated by Cell 9 of `New_BioMed_Scribe_Classification.ipynb`.

8.2 Cluster Structure

Compared to the single-block model (v2), the two-block model produces more uniformly distributed embeddings across the 2D t-SNE space. In v2, Neurology formed a very tight isolated cluster and Orthopedics separated sharply to the lower-left. In v3 the classes are more spread out while still showing partial separation. This reflects that the two-block model learned more nuanced, distributed representations rather than concentrating class information in a few dominant directions.

8.3 Embedding Magnitude

The mean L2 norm of hidden state vectors is more uniform across specialties in v3 than in v2. In v2 the norms ranged from 98 to 146, a spread of 48 units. In v3 they range from 84.8 to 105.4, a spread of only 21 units. This suggests the two-block model encodes specialty information primarily in the *direction* of the embedding vectors rather than their *magnitude*, which is generally a more

efficient use of the representation space and is consistent with the improved classification accuracy. Table 8 reports the per-class statistics.

Specialty	Mean L2 Norm	Std Dev
Cardiology	105.4	49.9
Neurology	93.5	44.4
Orthopedics	100.6	42.9
General Medicine	84.8	40.6

Table 8: Mean embedding magnitude (L2 norm) per specialty for 500 test examples. The two-block model produces more uniform norms than the single-block model (v2 range: 98–146; v3 range: 85–105).

9 Deployment

9.1 Gradio Web Application

The trained classifier is deployed as an interactive Gradio application. The interface accepts a free-text doctor-patient conversation, tokenises it to a maximum of 512 tokens, and returns the predicted specialty with a full confidence breakdown across all four classes. A low-confidence warning is displayed when the top prediction falls below 50% confidence, prompting the user to review the case manually.

9.2 Daily Startup Cell

A self-contained Colab cell loads the saved model from Google Drive and launches the Gradio UI without retraining. The cell mounts Drive, rebuilds the `GPT2MedicalClassifier` class with the two-block configuration, loads the frozen backbone from the saved checkpoint, loads the classifier head weights from `classifier_bundle.pt`, and launches the public Gradio URL. Total startup time is approximately 90 seconds.

9.3 Live Demo Results

Ten test conversations spanning all four specialties were classified with the following results:

The one incorrect prediction is clinically meaningful: a patient presenting with racing heart, skipping beats, and dizziness when standing genuinely could be referred to either Cardiology or Neurology in a real clinical setting. The 39% confidence score shows the model correctly flagging this ambiguity rather than committing incorrectly with high confidence.

10 Limitations and Future Work

Expected	Predicted	Confidence	Result
Cardiology	Cardiology	100%	✓
Neurology	Neurology	74%	✓
Orthopedics	Orthopedics	58%	✓
General Medicine	General Medicine	96%	✓
Neurology	Neurology	100%	✓
Orthopedics	Orthopedics	83%	✓
Cardiology	Neurology	39%	×
General Medicine	General Medicine	99%	✓
Orthopedics	Orthopedics	72%	✓
Neurology	Neurology	100%	✓
Total			9/10

Table 9: Live demo results across 10 unseen test conversations. The single incorrect prediction (row 7) involved heart palpitations with dizziness, and dizziness is a strong Neurology keyword and the model’s low confidence (39%) correctly signals uncertainty.

10.1 Current Limitations

Keyword-based labeling. Specialty labels are assigned algorithmically rather than by licensed clinicians. Keyword matching introduces labeling noise for presentations where symptom vocabulary overlaps across specialties, particularly at the boundary between General Medicine and Orthopedics. This is the primary source of remaining classification error.

Four-class scope. Real clinical triage involves many more specialties. Expanding the taxonomy to include Psychiatry, Dermatology, and Gastroenterology would require substantially more labeled examples and possibly a larger pre-trained model.

Dataset domain. ChatDoctor Q&A pairs and AGBonnet multi-turn conversations differ structurally from real hospital encounter transcripts. Performance on actual clinical dialogue may differ from the reported results.

No pre-training on clinical text. Models such as BioBERT [3] and ClinicalBERT continue pre-training on biomedical text before task-specific fine-tuning, which gives them a stronger prior on clinical language. Our approach relies on GPT-2’s general language representations to transfer to the clinical domain without this intermediate step.

10.2 Future Work

Expert annotation. Replacing keyword labeling with physician-provided labels for even 3,000 to 5,000 conversations would likely produce larger accuracy gains than any further architectural

change, since labeling quality is currently the primary bottleneck.

Additional fine-tuning blocks. The natural next experiment is to unfreeze three blocks ($h[21]$, $h[22]$, $h[23]$) and observe whether the additional 12.6M trainable parameters continue to improve accuracy or begin to overfit on the available training data.

Expanded specialty taxonomy. Adding Psychiatry, Dermatology, and Gastroenterology as additional classes would produce a more clinically representative triage system.

EHR evaluation. Validating the classifier on electronic health record transcripts from a clinical institution would provide a more honest estimate of real-world performance.

11 Conclusion

This project demonstrates that GPT-2 Medium can be adapted for medical specialty classification through parameter-efficient fine-tuning on modest labeled data. Following a suggestion from the course instructor, we unfroze the final two transformer blocks rather than only the last one, increasing trainable parameters from 12.6M to 25.2M (7.1% of total 354.8M). Training for 14 epochs on 10,793 labeled clinical conversations achieves 79.1% test accuracy and a macro F1 of 0.799, improving every per-class metric relative to both prior model versions. Neurology achieves a recall of 93.6%, the strongest single-class result across all versions.

The embedding analysis shows that two-block fine-tuning produces more uniformly distributed representations across specialties, with embedding magnitude varying less than in the single-block model. This suggests the model encodes specialty information in the direction rather than the magnitude of hidden state vectors, which is a more efficient use of representation space.

The confusion matrix confirms that remaining errors are clinically explainable, concentrated at the boundary between General Medicine and Orthopedics where symptom vocabulary genuinely overlaps. The live demo achieves 9 out of 10 correct predictions, with the one failure being a genuinely ambiguous case flagged correctly by a low confidence score of 39%.

Future work should prioritise expert annotation to replace keyword-based labeling, expansion to additional medical specialties, and evaluation on real electronic health record data. This work demonstrates that task formulation and label quality, rather than model scale alone, are the primary determinants of performance in clinical NLP classification tasks.

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